Group and Representation Theory

Chapter 4. Tensor Operators and the Wigner-Eckart Theorem

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The $\Delta^+ \to N\pi$ Process

Let's consider a hadron decay process $\Delta^+ \to N\pi$:

$$\Delta^+ \to p + \pi^0$$
$$\Delta^+ \to n + \pi^+$$

Particle	Δ^+	p	n	π^0	π^+
Isospin	3/2	1/2	1/2	1	1
I_3	+1/2	+1/2	-1/2	0	+1

Table 1: Isospin and I_3 value of the particles involved.

Strong interactions approximately preserve **isospin**

The $\Delta^+ \to N\pi$ Process

Now we use Clebsch-Gordan decomposition $\frac{1}{2} \otimes 1 = \frac{3}{2} \oplus \frac{1}{2}$, to decompose

$$\Delta^+ = \left|\frac{3}{2}, \frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}} \left|p\pi^0\right\rangle + \sqrt{\frac{1}{3}} \left|n\pi^+\right\rangle \tag{1}$$

Therefore

$$\frac{\Gamma(\Delta^+ \to p\pi^0)}{\Gamma(\Delta^+ \to n\pi^+)} = \frac{\left|\left\langle \Delta^+ \middle| p\pi^0 \right\rangle\right|^2}{\left|\langle \Delta^+ | n\pi^+ \rangle\right|^2} \simeq 2 \tag{2}$$

Conclusion : the relative decay ratio between to processes is approximately 2!
Our result agrees with experimental observations- so it is **right**, but **why?**

The $\Delta^+ \to N\pi$ Process

This is because the S-matrix is an **isospin scalar** operator, so

$$\left\langle \alpha, I', M' \right| S \left| \beta, I, M \right\rangle = \delta_{II'}\delta_{MM'}\mathcal{A}_{\alpha\beta} \tag{3}$$

$\mathcal{A}$ describes the whole dynamics, and **independent** to and M
Therefore the internal dynamics are totally same for these two processes!

... **But why?**

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Tensor Operators

Tensor Operators

A **tensor operator** is a set of operators transforms under commutation with generators of the Lie algebra like rep of the algebra.

A set of operators $O^s_l, (l = -s, \cdots, s)$ transforming under the spin-s representation of $SU(2)$ satisfies

$$[J_a, O^s_l] = O^s_{l'}[J^s_a]_{l'l} \tag{4}$$

J^s_a : spin-s rep of $SU(2)$ algebra

Tensor Operators

Example : position operator $\hat{r}$
Consider a particle in a spherically symmetric potential $V(r)$, with no spin

$$\begin{aligned} J_a = L_a &= \epsilon_{abc} r_b p_c \\ [J_a, r_b] &= \epsilon_{acd}[r_c p_d, r_b] = \epsilon_{acd} r_c(-i\delta_{bd}) \\ &= r_c[J_a^{\text{adj}}]_{cb} \end{aligned}$$

Since adjoint rep of $SU(2)$ is equivalent to spin-1 rep, $\hat{r}$ is a tensor operator

Tensor Operators

By a suitable change of basis, we can put $\hat{r}$ to transform under **irreps**

1. A set of operators Ω_x ($x = 1, \cdots, 2s+1$) that transforms under a rep D :

$$[J_a, \Omega_x] = \Omega_y [J_a^D]_{yx} \tag{5}$$

2. Assume that J_a^D is equivalent to spin-s irrep : $S^{-1} J_a^D S = J_a^s$
3. **Define** a new set of operators $O_l^s = \Omega_y [S]_{yl}$
4. The new set of operators will transform under commutation like spin-s irreps

Tensor Operators

The new set of operators O_l^s satisfy...

$$\begin{aligned}
[J_a, O_l^s] &= [J_a, \Omega_y][S]_{yl} \\
&= \Omega_z [J_a^D]_{zy} [S]_{yl} \\
&= \Omega_z \delta_{zz'} [J_a^D]_{z'y} [S]_{yl} \\
&= \Omega_z [S]_{zm} [S^{-1}]_{mz'} [J_a^D]_{z'y} [S]_{yl} \\
&= O_m^s [J_a^s]_{ml}
\end{aligned}$$

Especially, for $[J_3^s]_{ll'} = l\delta_{ll'}$,

$$[J_3, O_l^s] = lO_l^s$$

Tensor Operators

Example. One can find the similarity transformation that brings J_a^{adj} to irrep J_a^1 :

$$S = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 0 & 1 \\ -i & 0 & -i \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$

The Cartesian $\hat{r}$ is now changed as

$$\frac{1}{\sqrt{2}} \begin{pmatrix} -1 & -i & 0 \\ 0 & 0 & \sqrt{2} \\ 1 & -i & 0 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} = \begin{pmatrix} -\frac{\hat{x}+i\hat{y}}{\sqrt{2}} \\ \hat{z} \\ \frac{\hat{x}-i\hat{y}}{\sqrt{2}} \end{pmatrix} \equiv T_q^{(1)} \tag{6}$$

Transformed operators are called the **rank-1 tensor operators**

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The Wigner-Eckart Theorem

Now we want to find **matrix elements** of the tensor operator O_l^s

$$\left\langle J, m', \beta \right| O_l^s \left| j, m, \alpha \right\rangle \tag{7}$$

the tensor operator O_l^s may be dipole operator, S-matrix, etc

First, let's examine transformation properties of $O_l^s \left| j, m, \alpha \right\rangle$ under J_a

$$J_a O_l^s \left| j, m, \alpha \right\rangle \tag{8}$$

The Wigner-Eckart Theorem

$$J_a O_l^s \left|j, m, \alpha\right\rangle = \left[J_a, O_l^s\right] \left|j, m, \alpha\right\rangle + O_l^s J_a \left|j, m, \alpha\right\rangle \tag{9}$$

$$= O_{l'}^s \left|j, m, \alpha\right\rangle \left[J_a^s\right]_{l'l} + O_l^s \left|j, m', \alpha\right\rangle \left[J_a^j\right]_{m'm} \tag{10}$$

Especially,

$$J_3 O_l^s \left|j, m, \alpha\right\rangle = (l + m) O_l^s \left|j, m, \alpha\right\rangle \tag{11}$$

Therefore they transform according to the tensor product $s \otimes j$

$$J_a(\left|s, l\right\rangle \otimes \left|j, m\right\rangle) = (J_a^s \left|s, l\right\rangle) \otimes \left|j, m\right\rangle + \left|s, l\right\rangle \otimes (J_a^j \left|j, m\right\rangle)$$

The Wigner-Eckart Theorem

So we may apply **highest weight decomposition** to the state of tensor operators, to write

$$\sum_l O_l^s \left|j, M-l, \alpha\right\rangle \left\langle s, j, l, M-l | J, M\right\rangle = k_J \left|J, M\right\rangle \tag{12}$$

The coefficients $\langle s, j, l, M-l|J, M\rangle$ are **Clebsch-Gordan coefficients!**
The only difference is encoded in the constant k_J on the right-hand side

$$O_s^l \left|j, m, \alpha\right\rangle = \sum_{J=|j-s|}^{j+s} \left\langle J, l+m | s, j, l, m\right\rangle k_J \left|J, l+m\right\rangle \tag{13}$$

The Wigner-Eckart Theorem

Now the coefficient k_J is determined by another label α :

$$k_J \left|J, l+m, \alpha\right\rangle = \sum_{\beta} k_{\alpha\beta} \left|J, l+m, \beta\right\rangle \tag{14}$$

and the $k_{\alpha\beta}$ is called the **reduced matrix elements** (denoted $\mathcal{A}$)

$$\mathcal{A}_{\alpha\beta} = \left\langle J, \beta\right| O^s \left|j, \alpha\right\rangle \tag{15}$$

The whole dynamics are encoded in $\mathcal{A}_{\alpha\beta}$

The Wigner-Eckart Theorem

The Wigner-Eckart Theorem

The matrix elements of O_l^s are

$$\left\langle J, m', \beta \right| O_l^s \left| j, m, \alpha \right\rangle = \delta_{m', l+m} \left\langle J, l+m \right| s, j, l, m \rangle \mathcal{A}_{\alpha\beta}, \tag{16}$$

where $\mathcal{A}_{\alpha\beta} = \langle J, \beta | O^s | j, \alpha \rangle$.

$\mathcal{A}_{\alpha\beta}$ is **independent** of l, m - so if we compute $\mathcal{A}_{\alpha\beta}$ for one l, m, then we're done!

Example

Example. $\langle 1/2, 1/2, \alpha|\, r_3\, |1/2, 1/2, \beta\rangle = A$. What is $\langle 1/2, 1/2, \alpha|\, r_1\, |1/2, -1/2, \beta\rangle$?

First, decompose $r_1 = \frac{1}{2}(T_3^{(1)} - T_1^{(1)})$, where $T_q^{(1)}$ are defined in eqn (6). Since $r_3 = T_2^{(1)}$,

$$\langle 1/2, 1/2, \alpha|\, T_2^{(1)}\, |1/2, 1/2, \beta\rangle = \langle 1/2, 1/2|1, 1/2, 0, 1/2\rangle\, \mathcal{A} = -\frac{1}{\sqrt{3}}\mathcal{A} = A$$

By **selection rule** $\delta_{m',l+m}$,

$$\langle 1/2, 1/2, \alpha|\, r_1\, |1/2, -1/2, \beta\rangle = -\frac{1}{\sqrt{2}}\, \langle 1/2, 1/2, \alpha|\, T_1^{(1)}\, |1/2, -1/2, \beta\rangle \tag{17}$$

$$= -\frac{1}{\sqrt{2}}\, \langle 1/2, 1/2|1, 1/2, 1, -1/2\rangle\, \mathcal{A} = -\frac{1}{\sqrt{3}}\mathcal{A} = A! \tag{18}$$

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Products of Operators

Consider a product of two tensor operators : $O^{s_1}_{m_1} O^{s_2}_{m_2}$, then

$$[J_a, O^{s_1}_{m_1} O^{s_2}_{m_2}] = [J_a, O^{s_1}_{m_1}] O^{s_2}_{m_2} + O^{s_1}_{m_1} [J_a, O^{s_2}_{m_2}] \tag{19}$$

$$= O^{s_1}_{m_1'} O^{s_2}_{m_2} [D^{s_1}_a]_{m_1' m_1} + O^{s_1}_{m_1} O^{s_2}_{m_2'} [D^{s_2}_a]_{m_2' m_2} \tag{20}$$

so a product $O^{s_1}_{m_1} O^{s_2}_{m_2}$ transforms like $s_1 \otimes s_2$ rep, and can be decomposed!

$$[J_3, O^{s_1}_{m_1} O^{s_2}_{m_2}] = (m_1 + m_2) O^{s_1}_{m_1} O^{s_2}_{m_2} \tag{21}$$